Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub

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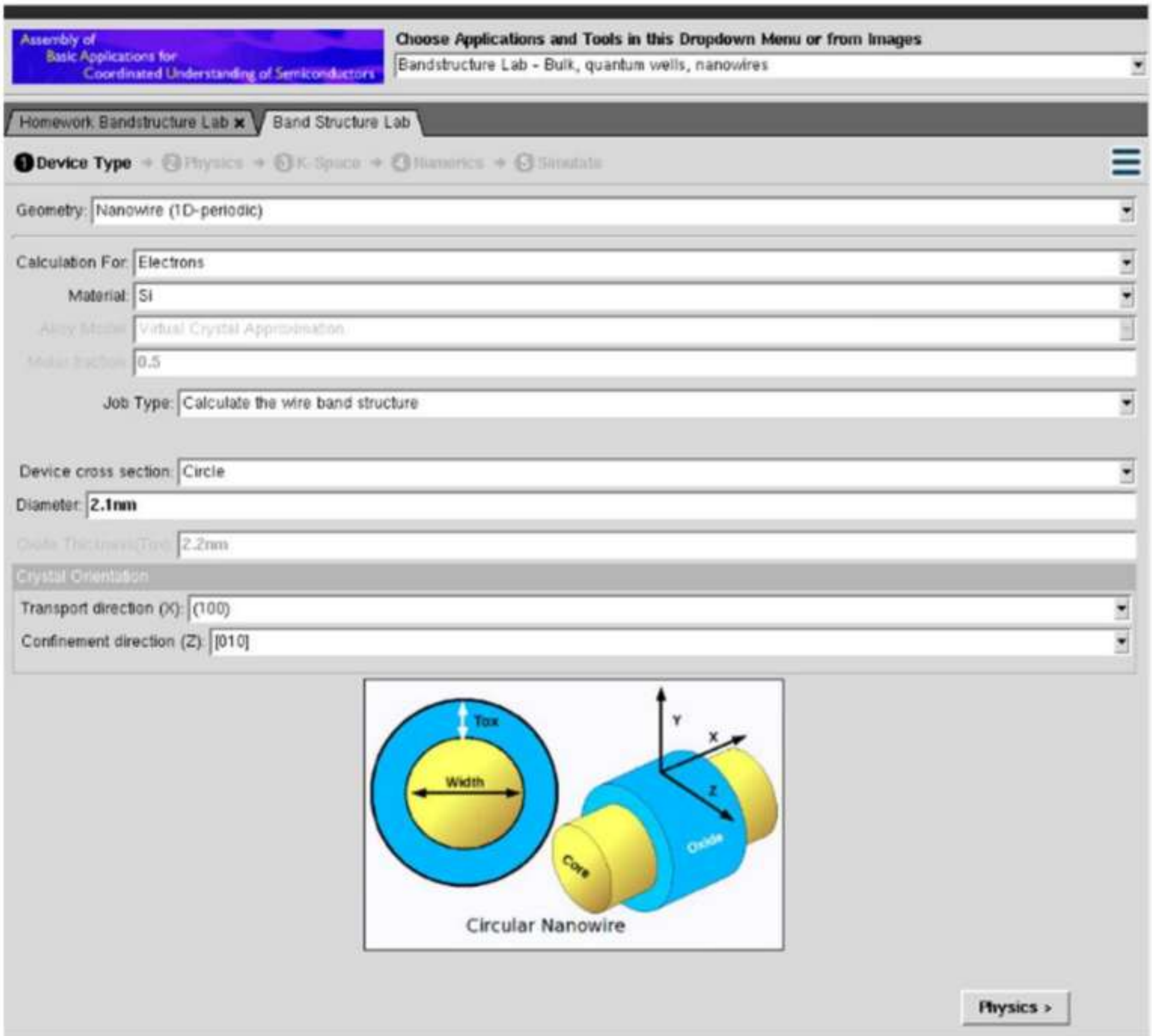
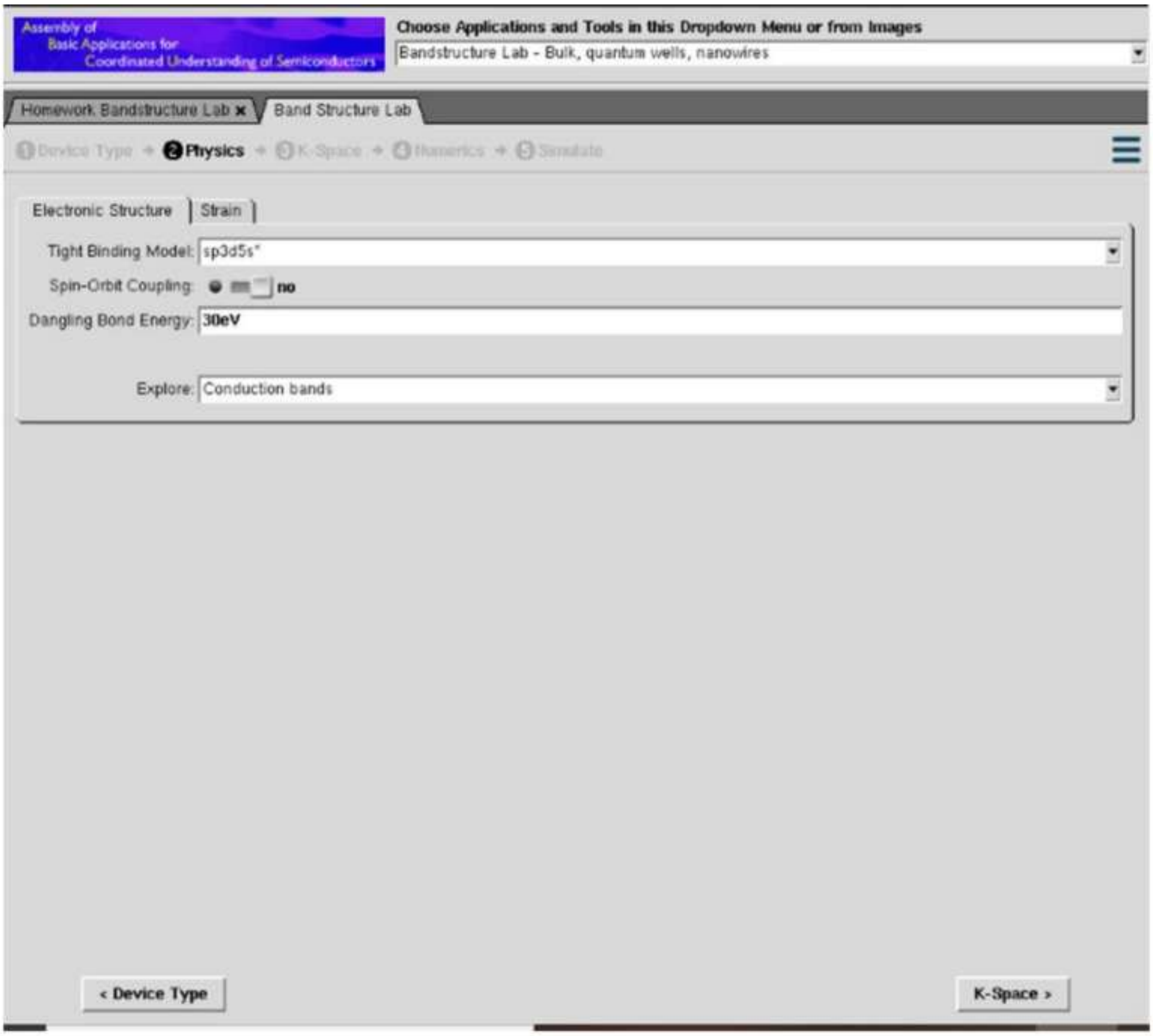
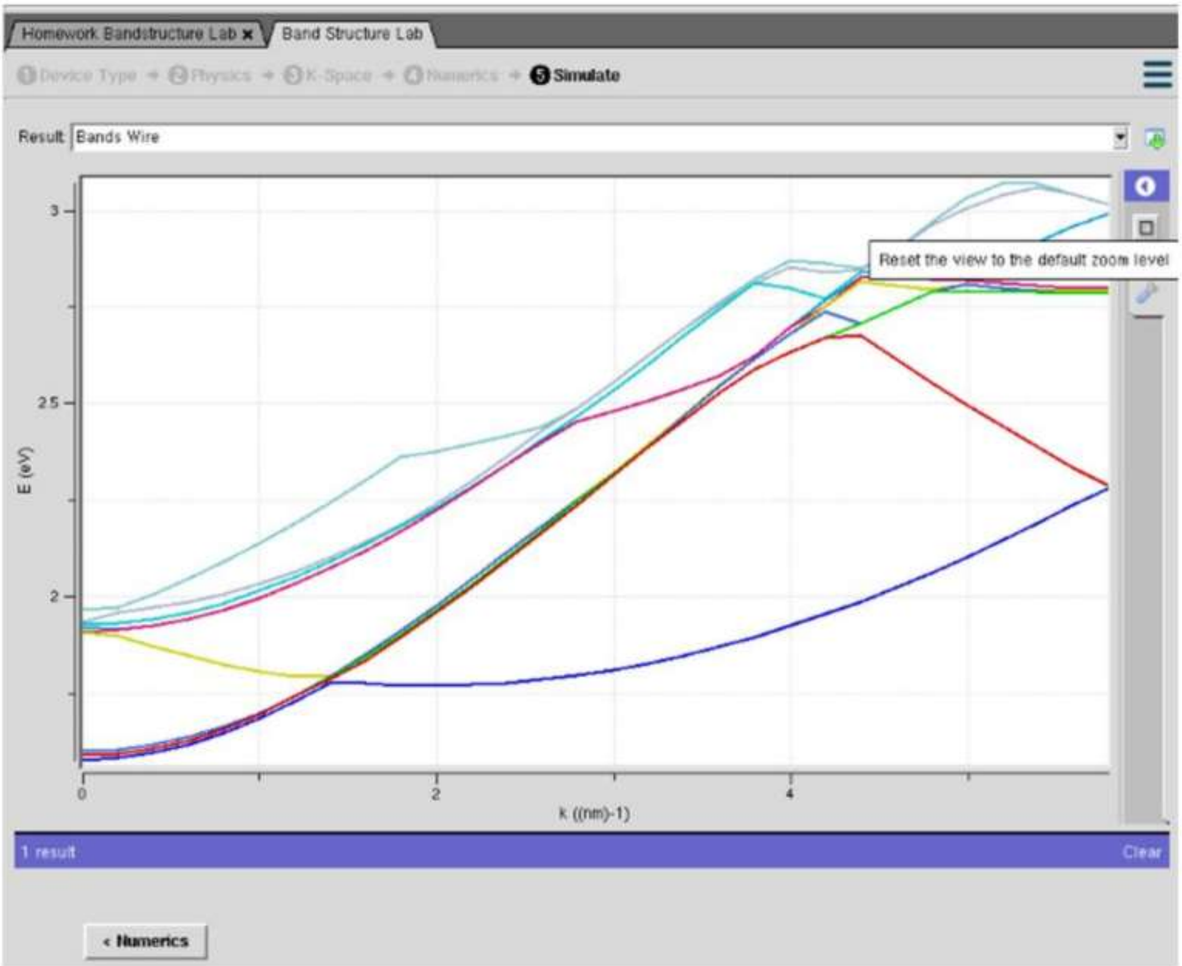
Slot-c ECA3102 NANO ELECTRONICS

OUTPUT

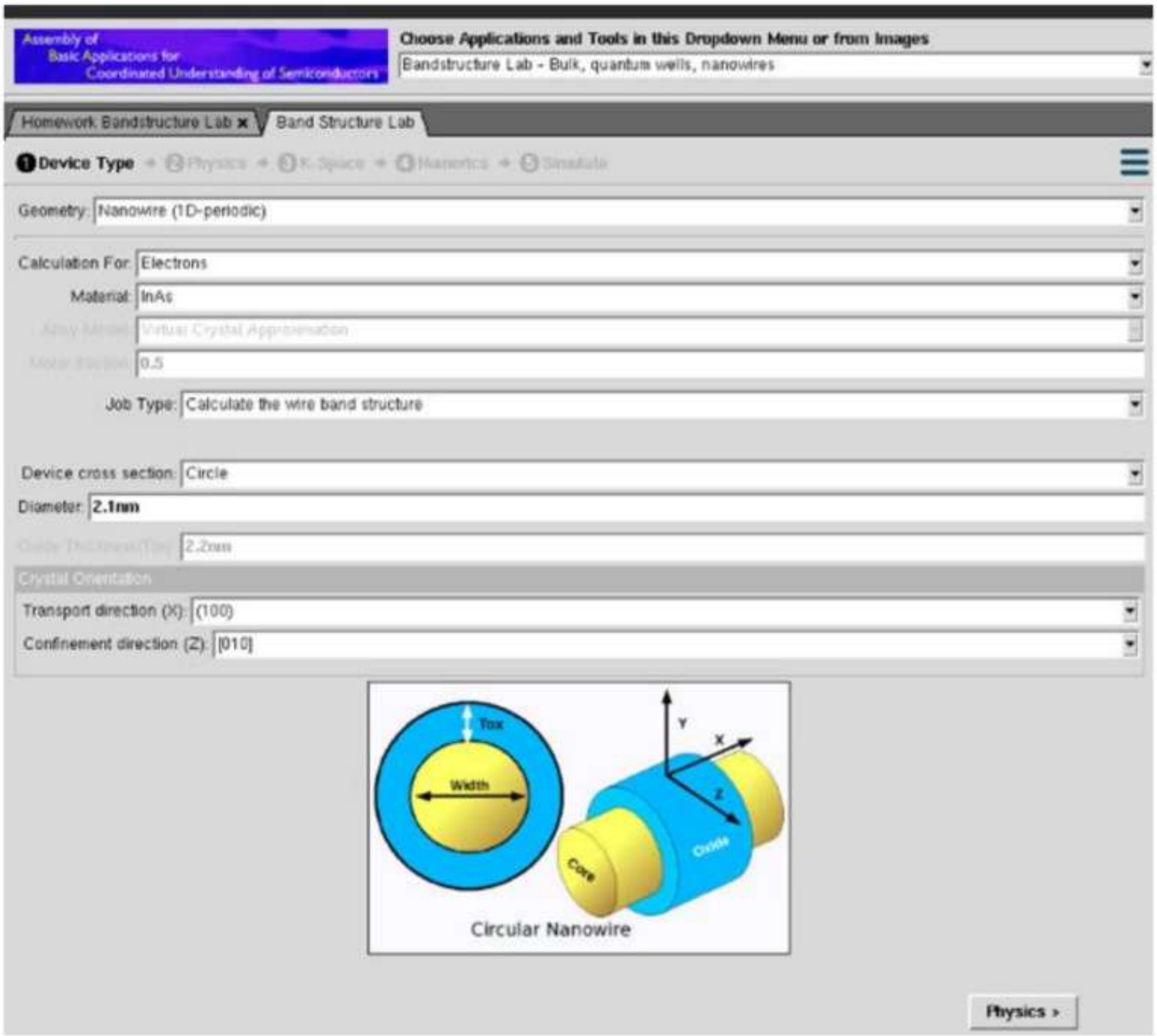


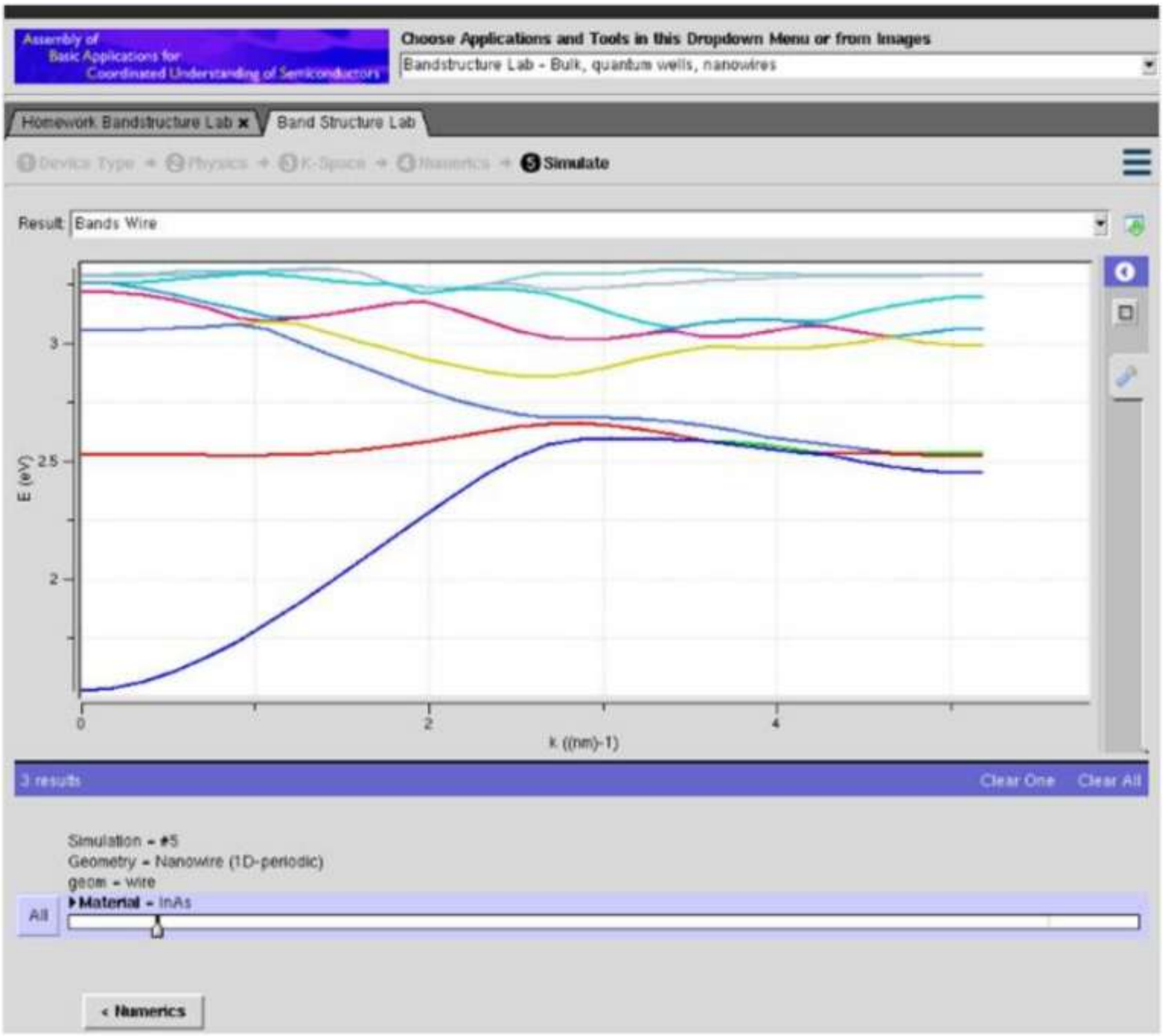

Assembly of
Basic Applications for
Coordinated Understanding of Semiconductors
Choose Applications and Tools in this Dropdown Menu or from Images
Bandstructure Lab - Bulk, quantum wells, nanowires
Homework Bandstructure Lab
Band Structure Lab
Device Type
Physics
K-Space
Numerics
Simulate
Electronic Structure
Strain
Tight Binding Model:
sp3d5s*
Spin-Orbit Coupling:
no
Dangling Bond Energy:
30eV
Explore:
Conduction bands
< Device Type
K-Space >



These figures show the simulation of a silicon nanowire band structure. The first image configures the nanowire geometry and material properties, the second sets the electronic structure parameters, and the third displays the band structure (Energy vs. Wave Vector), illustrating the allowed electron energy bands in the nanowire.

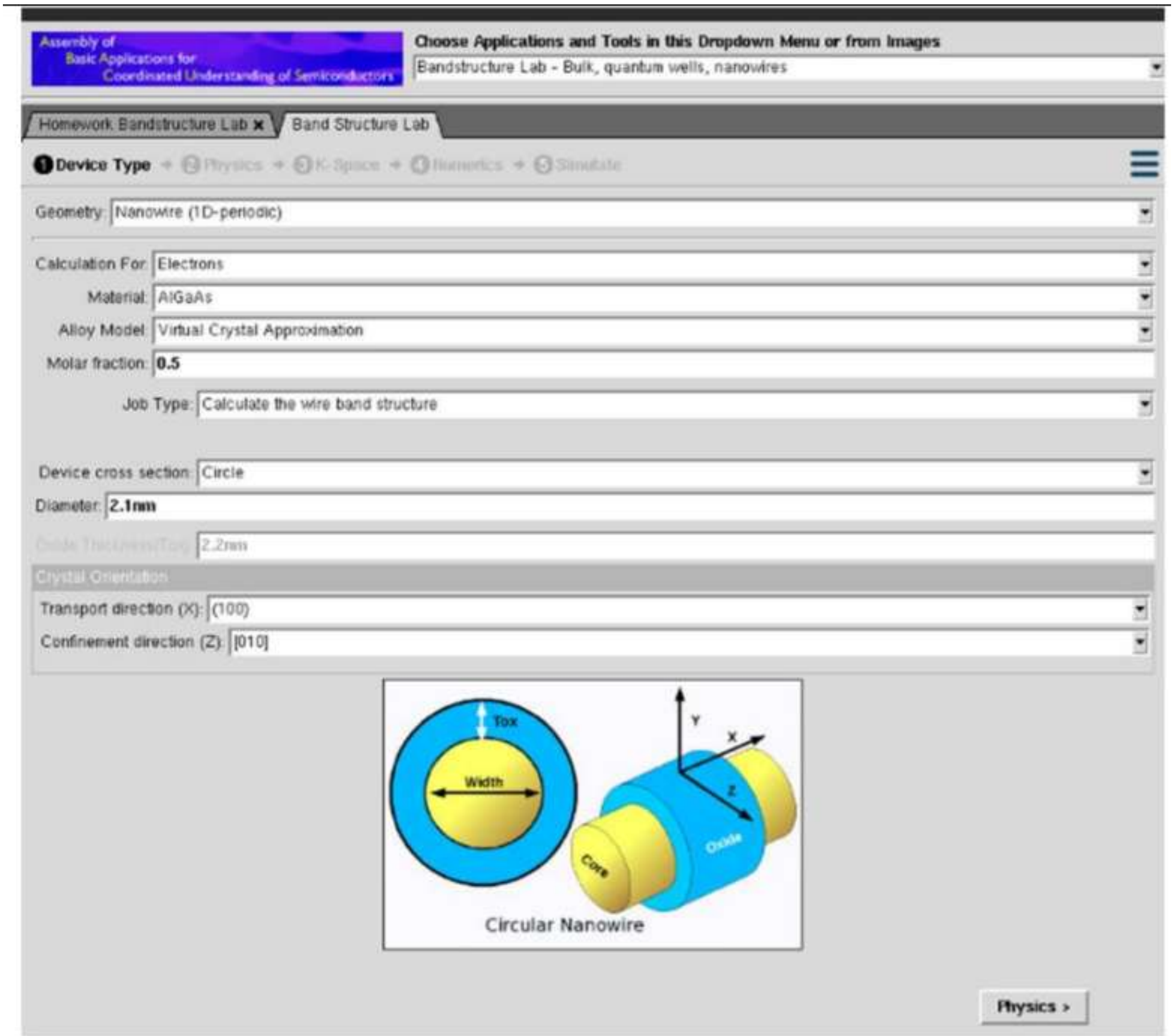
Case Study 1: Simulate the band-structure of InAs circular nanowire

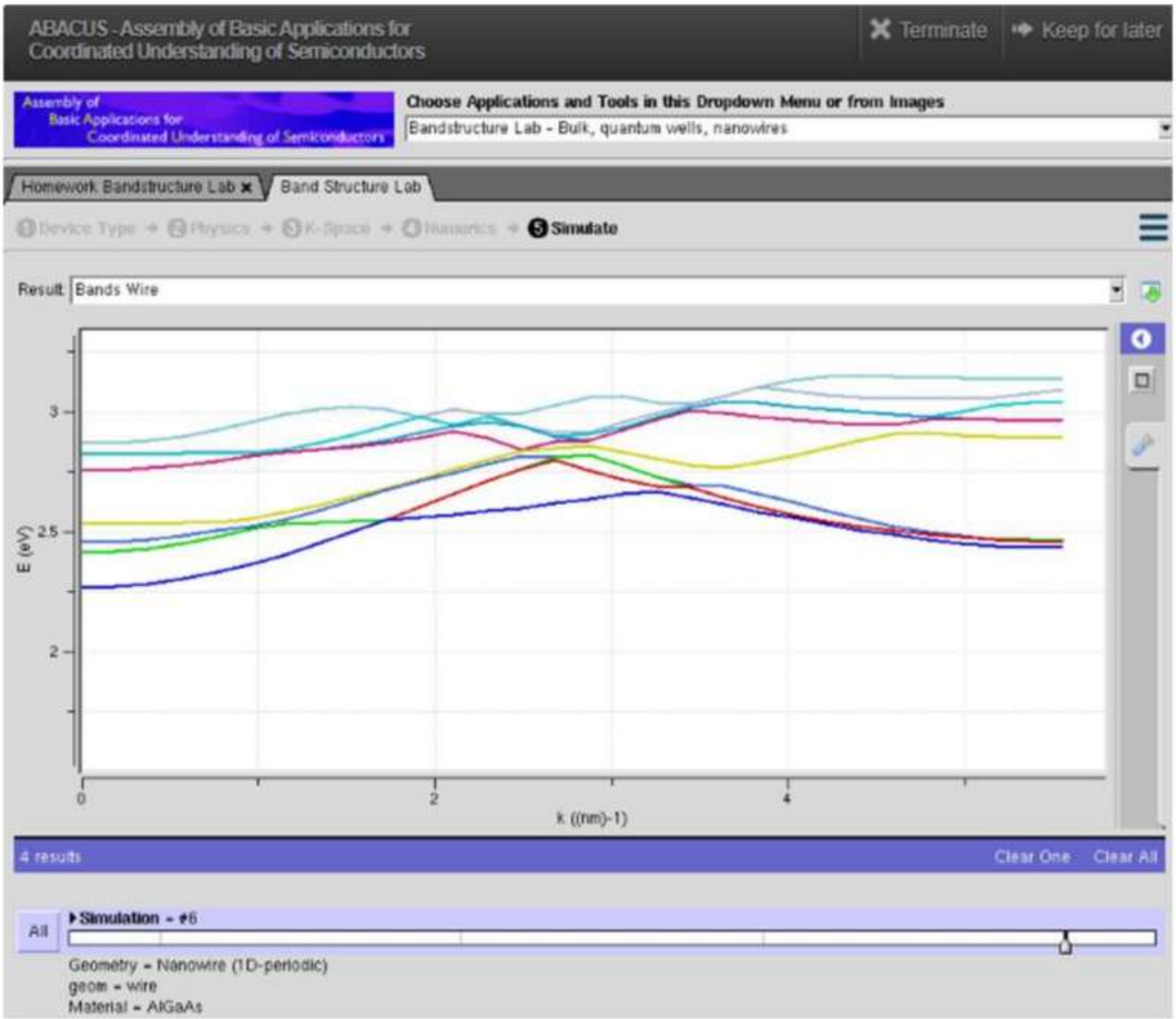




These images show the setup and simulation result of a nanowire band structure using semiconductor simulation software. The graph displays the energy bands of electrons, which helps analyze the nanowire's electrical properties.

Case Study 2: Simulate the band-structure of AlGaAs circular nanowire





These images show the setup and output of an AlGaAs nanowire band structure simulation.

The graph shows the electron energy bands, which are used to study the electrical properties of the nanowire.